MIL-S-63280(AR)
13 November 1978
SUPERSEDING
SAPD-216A
1 April 1966

MILITARY SPECIFICATION

SOLENOID
FOR MACHINE GUN, CALIBER .50: FIXED M85

This specification is approved for use by the U.S. Army Armament Research and Development, Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of electrical solenoid having a nominal operating voltage of 24 volts direct current used for remote firing of the M85 Machine Gun.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. - The following, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	-	Preservation, Methods of
MIL-W-13855	-	Weapons, Small Arms, General Specification for

STANDARDS

MILITARY

MIL-STD-100	-	Engineering Drawing Practices
MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	-	Quality Assurance Terms and Definitions

FSC: 5945

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-202	-	Test Methods for Electronic and Electrical Component Parts.
MIL-STD-1235	-	Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS (See 6.5)

U.S.ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

PRODUCT AND PACKAGING DRAWINGS

7792921	-	Solenoid
7792921	-	Solenoid - Packaging Data Sheet
8447859	-	Test Diagram and Requirements for Link, Cartridge, Metallic Belt, 20MM: M14A2

INSPECTION EQUIPMENT DRAWINGS

11015165	-	Gage, Test Set
11015609	-	Gage, Special Snap
11015611	-	Gage, Functional
11015612	-	Instructions, Operating for Gage 11015808
11015630	-	Gage, Instruction Sheet
11015806	-	Fixtrue, Operational Test
11015807	-	Gage, Functional
11015808	-	Gage, Indicator

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM E18	-	Standard Methods of Test for Rockwell Hardness and Rockwell Superfical Hardness of Metallic Materials

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(Application for copies should be addressed to: American Society for Testing and Material, 1916 Race Street, Philadelphia, PA 19103).

DEPARTMENT OF DEFENSE

DOD-D-1000 - Drawings, Engineering and Associated Lists.

3. REQUIREMENTS

3.1 Material and construction. - Solenoids and parts shall conform to the material and construction requirements specified herein, on Drawing D7792921 and drawings applicable thereto, and shall be in accordance with the applicable material and construction provisions of MIL-W-13855.

3.1.1 Potting. - The void space in the coil cavity shall be filled with potting compound as specified on the applicable drawing. The compound shall completely fill the holes in the sides of the block and shall be flush with the sides of the block.

3.1.2 Grounding of coil. - The coil shall be internally grounded as specified on the applicable drawing.

3.2 Components and assemblies. - Solenoids and parts shall comply with all requirments specified herein, on Drawings (Dwg.) 7792921, and associated drawings, and with all requirements specified in applicable specifications including MIL-W-13855.

3.2.1 Armature group. - The armature, with plunger assembly and disconnector pin attached, shall move through its full range of travel under spring action without binding of armature or plunger and shall be held normally and returned to the deenergized position by action of the spring.

3.2.1.1 Disconnector pin. - The disconnector pin shall be positioned in the angular location depicted on the applicable drawing. At each half turn within the adjustment range of the armature (See 3.2.1.3), the pin shall assemble and disassemble in the disconnector pin hole in the armature and shall be retained in the hole by engagement with the plunger assembly retaining detent.

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3.2.1.2 Retaining detent. - The retaining detent shall be retained in the plunger assembly by staking the plunger as specified on the applicable drawing. The retaining detent shall be capable of being depressed flush with or below the applicable surface of the plunger and when released shall return to its original position under spring action.

3.2.1.3 Plunger protrustion adjustment. - With the disconnector pin removed, the armature shall be capable of manual rotation in a counterclockwise or clockwise direction which shall respectivley advance or retract the plunger assembly. When plunger protrustion is set within the limits specified on the applicable drawing, the armature shall have an adjustment range from this setting of not less than 1 1/2 turns counterclockwise and one-half turn clockwise.

3.2.2 Pin, rivets, and sleeve bushing. - Dowel pins, rivets and riveted parts, and the sleeve bushing shall be tight and undamaged so that there shall be no relative movement.

3.2.3 Assembleability. - The solenoid shall readily assemble to and disassemble from a fixture simulating the back plate assembly of the M85 Machine Gun.

3.2.4 Operation. - Solenoids shall operate against a load of not less than 35 pounds at 18 and 30 volts direct current (d.c.) over the stroke range of 0.125 inch + 0.020 inch when tested as specified in 4.5.2.

3.2.5 Static force. - The minimum static force shall be 35 pounds over the 0.125 inch + 0.020 inch stroke at 18 volts d.c. when tested as specified in 4.5.3.

3.2.6 Coil resistance. - Coil resistance shall be 2.091 to 2.558 at 77° Fahrenheit (F) when tested as specified in 4.5.4.

3.2.7 Firing ability. - Solenoids shall be capable of firing, 1,000 rounds from an M85 Machine Gun without stoppages or malfunctions attributable to the solenoid when tested as specified in 4.5.5.

3.2.8 Reliability. - Solenoids shall be capable of passing a 10,000 cycle reliabitiy test without function failures or unserviceable parts when tested as specified in 4.5.6.

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3.2.9 Temperature rise. - Solenoids energized continuously for 5 minutes at a temperature of 160° F as specified in 4.5.7 shall be capable of producing a force of not less than 35 pounds throughout the test and, after being subjected to the temperature rise test, shall be capable of meeting the static force requirements specified in 3.2.5.

3.2.10 Temperature range. - Solenoids subjected to temperatures of minus 65° F. and plus 160° F. as specified in 4.5.8 shall be capable of meeting the static force requirments specified in 3.2.5.

3.2.11 Resistance to immersion. - Solenoids subjected to the immersion test specified in 4.5.9 shall be capable of meeting the static force requirments specified in 3.2.5.

3.2.12 Resistance to salt spray. - Solenoids subjected to the salt spray test specified in 4.5.10 shall be capable of meeting the static force requirments specified in 3.2.5.

3.2.13 Resistance to vibration. - Solenoids subjected to the vibration test specified in 4.5.11 shall be capable of meeting the static force requirments specified in 3.2.5.

3.2.14 Resistance to shock. - Solenoids subjected to the shock test specified in 4.5.12 shall be capable of meeting the static force requirements specified in 3.2.5.

3.2.15 Resistance to sand and dust. - Solenoids subjected to the sand and dust test specified in 4.5.13 shall be capable of meeting the static force requirements specified in 3.2.5.

3.3 Workmanship. - Workmanship shall be in accordance with the workmanship requirements of MIL-W-13855. In addtion the solenoid shall be free of dust, grease, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any parts nor shall the parts be contaminated by the cleaning agent. All marking and stamping shall be neat and clearly defined.

4. QUALITY ASSURANCE PROVISIONS

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4.1 Responsiblity for inspection. - Unless otherwise specified herein or in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. In addition, the Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define the terms used herein. The provisions of MIL-W-13855 shall apply.

4.2 Classification of inspections. - The following types of inspections shall be conducted on this item.

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First article inspection

4.3.1 Submission. - The contractor shall submit a first article as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantitites as indicated:

Part Description	Drawings	Quantity
Block	7792923	5
Bushing	7792924	5
Plate, bobbin	7792929	5
Plunger	7792939	5
Armature	7792942	5
Pin, disconnector	7792944	5
Plunger assembly	7792938	5
Solenoid	7792921	10

4.3.2 Inspections to be performed. - First article assemblies, components and test specimens may be subjected by the Government to any or all of the examinations and tests specified in Table I and to any or all of the requirements of the applicable drawings.

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4.3.3 Rejection. - If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any assembly, component, or test specimen in the sample to comply with any of the stated requirements.

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TABLE I - FIRST ARTICLE INSPECTIONS

CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 1 OF 3		DRAWING NUMBER
	SOLENOID AND COMPONENTS				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	SOLENOID				
	(Dwg 7792921)				
	Examination for defects	10[1]		3.2	4.4.2.8
	Coil Resistance	2		3.2.6	4.5.4
	Operation			3.2.4	4.5.2
	Static force			3.2.5	4.5.3
	Vibration			3.2.13	4.5.11
	Operation			3.2.4	4.5.2
	Shock			3.2.14	4.5.12
	Coil resistance			3.2.6	4.5.4
	Operation			3.2.4	4.5.2
	Static force			3.2.5	4.5.3
	Temperature range			3.2.10	4.5.8
	Coil resistance[2]	2		3.2.6	4.5.4
	Operation			3.2.4	4.5.2
	Static force			3.2.5	4.5.3
	Sand and Dust			3.2.15	4.5.13
	Operation			3.2.4	4.5.2
	Reliability			3.2.8	4.5.6
	Coil resistance			3.2.6	4.5.4
	Operation			3.2.4	4.5.2
	Static force			3.2.5	4.5.3

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 2 OF 3	DRAWING NUMBER
	SOLENOID AND COMPONENTS		See Below
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Coil resistance[2]	2		3.2.6	4.5.4
	Operation			3.2.4	4.5.2
	Static force			3.2.5	4.5.3
	Immersion			3.2.11	4.5.9
	Salt spray			3.2.12	4.5.10
	Coil resistance			3.2.6	4.5.4
	Operation			3.2.4	4.5.2
	Static force			3.2.5	4.5.3
	Temperature rise	2		3.2.9	4.5.7
	Firing ability[3]	2		3.2.7	4.5.5
	Block (Dwg 7792923)				
	Examination for defects	5		3.2	4.4.2.1
	Bushing (Dwg 7792924)				
	Examination for defects	5		3.2	4.4.2.2
	Plate, bobbin (Dwg 7792929)	5		3.2	4.4.2.3
	Plunger (Dwg 7792939)	5		3.2	4.4.2.4

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 3 OF 3	DRAWING NUMBER
	SOLENOID AND COMPONENTS		See Below
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Armature (Dwg. 7792942)	5		3.2	4.4.2.5
	Pin, Disconnector (Dwg. 7792944)	5		3.2	4.4.2.6
	Plunger assembly (Dwg 7792938)	5		3.2	4.4.2.7

NOTES: [1]Sample of (10) solenoids shall be numbered 1 through 10 in a manner that will permit positive identification throughout testing.
[2]Tests to be conducted in sequence listed
[3]To be conducted by the Government.

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4.4 QUALITY CONFORMANCE INSPECTION

4.4.1 Inspection lot formation. - The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions and complies with the provisions for submission of product as specified in MIL-STD-105. All material submitted for inspection in accordance with the specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examination. - The examinations listed in this specification for Classification of Defects and Tests shall be performed on inspection lots as defined in 4.4.1 of this specification.

a. Critical and special defects. - Unless otherwise specified, inspection for these defects shall be 100 percent. For defintions, see 6.4.3 and 6.4.4 of this specification.

b. Major and minor defects. - Major and minor defects shall be inspected by sampling to the AQL indicated in the Classification of Defects and Tests paragraphs provided in this specification, unless 100 percent inspection is indicated.

c. Sampling plans. - Unless otherwise specified in the classification of Defects and Test Tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105. In cases where sampling is specified in accordance with MIL-STD-105, the contractor may request permission from the procuring activity to use an equivalent continuous sampling plan from MIL-STD-1235.

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QUALITY CONFORMANCE INSPECTION

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 7792923
4.4.2.1	Block				NEXT HIGHER ASSEMBLY 7792922
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101.	Angle and depth of milled slot (2 places)		0.65%	3.2	11015609
102.	Depth of screw counterbore, min. (2 Places)		0.65%	3.2	Gage
103.	Overall thickness		0.65%	3.2	Gage
Minor					
201.	Thickness of bolt clearance slot, max.		1.5%	3.2	Gage
202.	Evidence of poor workmanship		2.5%	3.3	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 7792924
4.4.2.2	Bushing				NEXT HIGHER ASSEMBLY 7792922
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101.	Inside diameter		0.65%	3.2	Gage
Minor					
201.	Evidence of poor workmanship		2.5%	3.3	Visual
202.	Largest outside diameter, max.		1.5%	3.2	Gage
203.	Length of largest outside diameter, max.		1.5%	3.2	Gage

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 7792929
4.4.2.3	Plate, Bobbin				NEXT HIGHER ASSEMBLY 7792946
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101.	Overall thickness		0.65%	3.2	Gage
Minor					
201.	Thickness of mounting hole surface (2 places)		1.5%	3.2	Gage
202.	Width of center section		1.5%	3.2	Gage
203.	Distance from center to side of center section		1.5%	3.2	Gage
204.	Depth of clearance counterbore for sleeve bushing, min.		1.5%	3.2	Gage
205.	Diameter of clearance counterbore for sleeve bushing, min.		1.5%	3.2	Gage
206.	Evidence of poor workmanship		2.5%	3.3	Visual

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 7792939
4.4.2.4	PLUNGER				NEXT HIGHER ASSEMBLY 7792938
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201.	Radius at end		1.5%	3.2	Gage
202.	Evidence of poor workmanship		2.5%	3.3	Visual

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PARAGRAPH	TITLE		SHEET	OF	DRAWING NUMBER
4.4.2.5	ARMATURE		1	1	7792942
					NEXT HIGHER ASSEMBLY 7792941
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201.	Depth of screwdriver slot		1.5%	3.2	Gage
202.	Width of screwdriver slot		1.5%	3.2	Gage
203.	Evidence of poor workmanship		2.5%	3.3	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63280(AR)

PARAGRAPH	TITLE			SHEET 1 OF 1	DRAWING NUMBER 7792944
4.4.2.6	PIN, DISCONNECTOR				NEXT HIGHER ASSEMBLY 7792921
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101.	Diameter		.65%	3.2	Gage
102.	Depth of groove		.65%	3.2	Gage
103.	Radius of groove		.65%	3.2	Gage
104.	Hardness		.65%	3.2	4.5.16
Minor					
201.	Evidence of poor workmanship		2.5%	3.3	Visual

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 7792938
4.4.2.7	PLUNGER ASSEMBLY				NEXT HIGHER ASSEMBLY 7792921
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101.	Stake missing or improper		.65%	3.2	Visual
102.	Plunger does not function properly		.65%	3.2.1.2	Manual
Minor					
201.	Evidence of poor workmanship		2.5%	3.3	Visual

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE			SHEET 1 OF 2	DRAWING NUMBER 7792921
4.4.2.8	Solenoid				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101.	Coil resistance		100%[3]	3.2.6	4.5.4
102.	Static force		100%[3]	3.2.5	4.5.3
103.	Assembleability		.65	3.2.3	11015807
104.	Disconnector pin does not assemble/disassemble properly		.65	3.2.1.1	Manual
105.	Plunger protrusion and stroke		.65	3.2	11015808, 11015612
106.	Height location and centrality of pin		.65	3.2	11015611
107.	Improper functioning of armature and plunger		.65	3.2.1	Manual
108.	Plunger protrusion adjust improper		.65	3.2.1.3	Manual
109.	Potting compound not flush with sides of block		.65	3.1.1	Gage
110.	Missing or improperly assembled parts		.65	3.2	Visual/Manual
111.	Operation		.65[1,3]	3.2.4	11015806
112.	Temperature rise		.65[1,3]	3.2.9	11015806
113.	Temperature rise	1[2]	ACC-0 REJ-1	3.2.9	11015806
114.	Temperature range	1[2]	ACC-0 REJ-1	3.2.10	11015806
115.	Reliability	1	ACC-0 REJ-1	3.2.8	11015806
116.	Vibration	1	ACC-0 REJ-1	3.2.13	11015806

NOTES:

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PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.8	SOLENOID				7792921
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201.	Marking missing, incorrect or illegible		1.5	3.2	Visual
202.	Evidence of poor workmanship		2.5	3.3	Visual

NOTE 1 Inspection level S-3 of MIL-STD-105.
2 The same sample shall be subjected to the temperature rise than to the temperature range test.
3 The sequence of there tests shall be: operation; coil resistance; static force; temperature rise.

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PARAGRAPH	TITLE		SHEET OF		DRAWING NUMBER
4.4.2.9	SOLENOID - PACKAGING DATA SHEET		1 1		7792921
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None				
Major					
101.	Leakage test	*	*	3.2	4.5.14
102.	Heat seal test	*	*	3.2	4.5.15
Minor					
201.	Solenoid improperly cleaned		1.5	3.2	Visual
202.	Wrap not as specified		1.5	3.2	Visual
203.	Cushioning not as specified		1.5%	3.2	Visual
204.	Improper amount of desiccant		1.5%	3.2	Visual
205.	Improper sealing of unit container		1.5	3.2	Visual
206.	Intermediate container not as specified		1.5	3.2	Visual
207.	Improper closure of intermediate container		1.5	3.2	Visual
208.	Shipping container not as specified		1.5	3.2	Visual
209.	Improper closure of shipping container		1.5	3.2	Visual
210.	Marking missing, incorrect or illegible		1.5	3.2	Viusal

NOTES: *Sampling and rejection shall be in accordance with MIL-P-116.

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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4.4.3 Inspection equipment. - The inspection equipment required to perform the examinations and test prescribed herein is described in the Paragraph Reference/Inspection Method column in the tables starting with 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Sections 6.2 and 6.3 herein.

4.5 Test methods and procedures

4.5.1 Test conditions and power supply. - Unless otherwise specified, solenoids shall be tested at an ambient termperature of 77° ± 7°F. The power supply used to test the solenoids shall be capable of delivering 15 amperes over a range of 18 to 30 volts d.c.

4.5.2 Operation test. - Solenoids shall be tested for the operation requirement (See 3.2.4) using a test fixture conforming to Drawing 11015806. The solenoid shall be mounted on the test fixture and shall be operated against the specified load with an applied voltage of 18.0 to 18.5 volts d.c. and 29.5 to 30.0 volts d.c. The solenoid shall be cycled, 0.6 seconds on and 10 seconds off, for 45 minutes at each voltage. The test fixture shall be examined during the test to determine that the solenoid is producing the required travel.

4.5.3 Static force test. - Solenoids shall be tested for the static force requirement (See 3.2.5) using a test fixture conforming to Drawing 11015806. The solenoid shall be mounted on the test fixture and shall be operated against the specified load three times with an applied voltage of 18.0 to 18.5 volts d.c. The test fixture shall be examined during the test to determine that the solenoid is producing the required travel.

4.5.4 Coil resistance test

4.5.4.1 Solenoids shall be tested for the coil resistance requirement (See 3.2.6) in accordance with instructions specified on Drawing 11015630 using a test set conforming to Drawing 11015165.

4.5.4.2 In lieu of the 77°F temperature specified in 3.2.6 solenoids may be tested at other temperatures with suitable correlations in the resistance requirements as indicated on Drawing 11015630.

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4.5.5 Firing ability test. - Solenoids shall be tested for the firing ability requirement (See 3.2.7) by assembling the solenoids to an M85 machine gun of known acceptablity and firing 1,000 rounds of ammunition and cycling the solenoid 4 seconds on and 8 seconds off. Five hundred rounds shall be fired with the gun set at the low rate of fire and 500 rounds shall be fired at the high rate of fire. The machine gun shall be cooled for 10 minutes by forced air after the firing of 500 rounds at the low rate and after firing the first 250 rounds at the high rate. The voltage applied at the solenoid shall be 18.0 to 18.5 volts d.c. This is a destructive test. Assemblies so tested shall not be returned to the lot.

4.5.6 Reliability test. - Solenoids shall be tested for the reliability requirement (See 3.2.8) using a test fixture conforming to Drawing 11015806. The solenoid shall be mounted on the test fixture and operated through 10,000 cycles, 0.6 seconds on and 10 seconds off, against a 35-pound load with an applied voltage of 24.0 $\pm$ 2.0 volts d.c. The test fixture shall be examined during the test to determine the solenoid is producing the required travel. This is a destructive test. Assemblies so tested shall not be returned to the lot.

4.5.7 Temperature rise test. - Solenoids shall be tested for the temperature rise requirement (See 3.2.9) using a test fixture conforming to Drawing 11015806. The solenoid shall be mounted on the test fixture and the solenoid and test fixture shall be placed in a temperature test cabinet. The solenoid shall be energized continuously against a 35-pound load for 5 minutes at a temperature of 160° $\pm$ 5° F. with an applied voltage of 24.0 $\pm$ 2.0 volts d.c. The test fixture shall be examined during the test to determine the solenoid is producing the required travel. At the end of 5 minutes, the solenoid shall be allowed to cool to room temperature and shall be subjected to the statice force test specified in 4.5.3.

4.5.8 Temperature range test. - Solenoids shall be tested for the temperature range requirement (See 3.2.10) using a test fixture conforming to Drawing 11015806. The solenoid shall be mounted on the test fixture, and the solenoid and test fixture shall be placed in a temperature test cabinet with electrical circuits connected as in intended operation. This is a destructive test. Assemblies so tested shall not be returned to the lot.

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4.5.8.1 Low temperature. - The solenoid shall be subjected to a temperature of minus 65° ± 5° F. for 24 hours. After the 24-hour period and while at this temperature, the solenoid shall be subjected to the static force test specified in 4.5.3.

4.5.8.2 High temperature. - After being allowed to return to room temperature, the solenoid shall be subjected to a temperature of 160° ± 5° F. for 24 hours. After the 24-hour period and while at this temperature, the solenoid shall be subjected to the static force test specified in 4.5.3.

4.5.9 Resistance to immersion test. - Solenoids shall be tested for the resistance to immersion requirement (See 3.2.11) in accordance with method 104, test condition B, of MIL-STD-202. Not less than 4 hours and not more that 24 hours after completion of the final immersion cycle, the solenoid shall be subjected to the static force test specified in 4.5.3. This is a destructive test. Assemblies so tested shall not be returned to the lot.

4.5.10 Resistance to salt spray test. - Solenoids shall be tested for theresistance to salt spray requirement (See 3.2.12) in accordance with method 101 of MIL-STD-202, except that the length of the test shall be 200 hours. After exposure to salt spray, the solenoid shall be subjected to the static force test specified in 4.5.3. This is a destructive test. Assemblies so tested shall not be returned to the lot.

4.5.11 Resistance to vibration test. - Solenoids shall be tested for the resistance to vibration requirement (See 3.2.13) in accordance with method 201 of MIL-STD-202. Mounting shall be as in intended operation and motion shall be in the direction of each of the three major axes for 1 hour in each axes. Upon completion of the three hours of vibration, the solenoid shall be subjected to the static force test specified in 4.5.3. This is a destructive test. Assemblies so tested shall not be returned to the lot.

4.5.12 Resistance to shock test. - Solenoids shall be tested for the resistance to shock requirement (See 3.2.14) in accordance with method 202 of MIL-STD-202. Mounting shall be as in intended operation and accelerating force shall be applied in the direction of each of the three major. The pulse duration shall be 7 ± 1 milliseconds. Upon completion of the shock, the solenoid shall be subjected to the static force test specified in 4.5.3. This is a destructive test. Assemblies so tested shall not be returned to the lot.

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4.5.13 Resistance to sand and dust test. - Solenoids shall be tested for the resistance to sand and dust requirement (See 3.2.15) in accordance with method 101, test condition A, of MIL-STD-202, except that the sand and dust velocity through the chamber shall be 2,300 ± 500 feet per minute. Upon completion of the test, the solenoid shall be subjected to the static force test specified in 4.5.3. This is a destructive test. Assemblies so tested shall not be returned to the lot.

4.5.14 Leakage test. - The leakage test shall be conducted in accordance with MIL-P-116.

4.5.15 Heat seat test. - The heat seat test shall be conducted in accordance with MIL-P-116.

4.5.16 Hardness. - Hardness testing shall be conducted in accordance with ASTM-E-18.

5. PACKAGING

5.1 Preservation and packaging.

5.1.1 Level A. - The solenoid shall be preserved and packaged for shipment in accordance with packaging data sheet 7792921.

5.2 Level A. - The solenoid shall be packed for shipment in accordance with packaging data sheet 7792921.

5.3 Marking. - Marking shall be in accordance with MIL-STD-129.

5.4 Shipping. - When shipments from more than one lot are shipped as a carload, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Ordering data. - Procurement documents shall specifiy the following:

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6.1.1 Procurement requirements:

a. Title, number and date of this and the item detail specifications.

b. Quantity required and delivery schedules.

c. Serialization requirements, if applicable.

d. Quality Conformance Inspection, if other than specified in Section 4 of item detail specification.

e. First Article Sample requirements, if other than specified in Section 4 of item detail specifcation.

f. Packaging requirements, if other than specified in Section 5 of item detail specification.

g. Certificates of conformance for each lot or shipment or product.

6.1.2 Contract Data Requirements

a. Contract data requirements for inspection equipment designs (conforming to Data Item Description DI-E-118 or DI-R-1714 with appropriate addendum to each). See 6.2.

6.2 Inspection equipment designs. - Inspection equipment deigns are of two types - Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers under the "Paragraph Reference/Inspection Method" heading in Section 4. Design responsiblity for all other inspection equipment is assigned to the contractor. However, the contractor need not furnish any design when a complete Government SIE design is part of the Technical Data Package (TDP). Unless otherwise specified, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.2.2 and 6.2.3 should he elect to do so.

6.2.1 SIE designs. - SIE designs may consist of any of the following:

a. Detail draiwngs which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

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b. A source control drawing or specification control drawing as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detailed design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.2.2 Contractor designs. - Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commerical equipment which the contractor proposes to use. (Commerical equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public). Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commerical equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commerical equipment, or with SIE designs if not otherwise covered thereby (See 6.2.1c). Designs shall be of the category and form (per DOD-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423).

The specification number, paragraph number, and defect number from Section 4 shall be referenced on each contractor design together with the component or assembly drawing number, revision letter and date to which the specific design applies.

6.2.3 Submission of design for approval. - Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements Lists (DD Form 1423) or the contract. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs and the required delivery schedule as stipulated in the contract. The specific segment of ARRADCOM to which the contractor designs shall be sent is specified in 6.3.

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6.3 Submission of inspection equipment designs for approval. - Submit equipment designs as required to Commander, ARRADCOM, Dover, N.J. 07801, ATTN: DRDAR-QAA-I. Letter of submittal should state: contractor, contract number, specification number, item nomenclature, classification of defects or test paragraph, and designs remaining to be submitted and the expected date of submittal.

6.4 Definitions

6.4.1 Contractor. - Unless otherwise specified, the word Contractor used throughout this or the item detail specification shall mean any supplier or producer of items or material to the Government. This shall include commerical contractors, subcontractors, Government Owned Contractor Operated (GOCO), and Government Owned Government Operated (GOGO) plants.

6.4.2 Technical data package (TDP). - A technical description of an item adequate for use in procurement. The description defines the required design configuration and assures adequacy of item performance. It consists of all applicable technical data such as plans, drawings and associated lists, specifications, standards, models, performance requirements, quality assurance provisions, and packaging data.

6.4.3 Critical defect. - A defect that judgement and experience indicate is likely to result in hazardous or unsafe conditions for individuals using, maintaining, or depending upon the product; or a defect that judgement and experience indicate is likey to prevent performance of the tactical function of a major end item such as an aircraft, tank, land vehicle, missile, artillery, or other major weapon system.

6.4.4 Special defect. - A defect, other than Critical, that judgement and experience indicate may, depending upon the degree of variance from the design requirement:

a. Result in hazardous or unsafe conditions for individuals using, maintaining or depending upon the product or

b. Prevent performance of the tactical function of a major end item.

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6.5 Drawings. - Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

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Activity
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Preparing
Army-AR

Project Number: 5945-A394

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